CERTAIN-COAST - concept note

Certified early warning to protect water, port and energy infrastructure on a sinking Adriatic coast

NATO Science for Peace and Security (SPS) Programme - Multi-Year Project, call 2027-1 - 36 months - about EUR 382,500 requested from NATO - coordinator: Faculty of Geology and Mining, Polytechnic University of Tirana, Albania

1. In short

Near Durrës, Albania's main port, the wells that supply drinking water, the port itself and an energy connection from the sea stand on low ground that is slowly sinking, close to active faults. Three dangers threaten them: seawater entering the wells, ground that turns almost liquid in an earthquake, and the sea flooding the land in storms. The dangers are linked, because sinking land and rising seas worsen all three and pumping that protects against one can increase another, but today they are watched separately and no warning says how often it can be wrong.

CERTAIN-COAST will watch them together. Satellites, GNSS and new ground sensors will feed one computer model of the coast, copied by a fast artificial-intelligence model whose error is stated for every forecast. Each warning will state in advance how often it may miss a dangerous event, and this will be checked in twelve months of live operation with the water utility, the port, the energy operator and civil protection. The project will also show how many years a safe way of operating the three assets together remains, and sharpen that estimate with short, safe pumping tests. Local operators will run the system afterwards.

2. The security challenge

Durrës, on the Adriatic coast of Albania, concentrates three kinds of critical infrastructure of a NATO Ally within a few kilometres: the wellfields that supply drinking water, the Port of Durrës, from which 93.6% of Albania's seaborne freight left in the second quarter of 2025 (INSTAT), and an energy landfall where supply arrives from the sea. All three stand on soft coastal sediments that are sinking, close to active thrust faults, and the Mw 6.4 earthquake of 26 November 2019 caused liquefaction along this coast.

The assets face three linked hazards. Seawater intrudes into the aquifer and can make wells unusable for years. In an earthquake, saturated sand can liquefy under quays, foundations and the landfall. Storm surges and extreme sea levels flood low ground. Land subsidence and sea-level rise worsen all three at once, because they raise the sea and the water table together. Pumping moves the hazards in opposite directions: it draws seawater towards the wells but lowers the water table and so strengthens the ground against liquefaction. Managing one hazard alone can increase another.

Today these hazards are monitored by different institutions with different models, and no warning states how often it misses. For security decisions this is the critical gap: an operator who must cut pumping, restrict a quay or inspect the landfall needs to know how far a warning can be trusted, and civil protection needs one consistent picture of water, port and energy after an earthquake or a storm.

The security challenge is therefore to keep three essential services of an Ally working, and to know in advance when they will stop being safe. The project serves three of NATO's seven baseline requirements for national resilience (resilient food and water resources, resilient civil transportation systems and resilient energy supplies) and the topic of particular interest of the 2027-1 call, "protection of critical infrastructure, supplies and personnel; maritime and undersea infrastructure resilience". NATO's 2022 Strategic Concept calls climate change "a crisis and threat multiplier"; on this coast the multiplication is physical and measurable.

A second challenge is trust in machine-learned tools used in security decisions. CERTAIN-COAST uses artificial intelligence only with a certified error bound and a miss rate that is checked in operation, in line with NATO's Principles of Responsible Use of AI.

3. Who uses it, and how

Link to security. The project protects three essential services of a NATO Ally (drinking water, a port that handles most of the country's seaborne freight, and an energy landfall) and serves three of NATO's seven baseline requirements for national resilience: resilient food and water resources, resilient civil transportation systems and resilient energy supplies, which also underpin civil support to

the military. It falls under the 2027-1 topic "protection of critical infrastructure, supplies and personnel; maritime and undersea infrastructure resilience", and the energy landfall is where undersea infrastructure comes ashore.

Use cases:

- Water utility: cut or shift pumping before chloride reaches drinking-water wells (weeks to months ahead).
- Port authority: restrict loads or operations and set inspection priorities after an earthquake; prepare for storm surges (hours to days ahead).
- Energy landfall operator: inspect, protect or isolate the landfall when the ground or the water level threatens it.
- Civil protection: one consistent bulletin for water, port and energy after an earthquake or a storm.
- National authorities and resilience planners: the certified year from which no operating plan keeps the three assets safe together, and the sensors and measures that delay it most.

Each bulletin states its certified miss rate, so decision-makers know how far it can be trusted. The method uses free satellite data and transfers to other subsiding coasts of Allies and partners with wellfields, ports and energy landfalls.

4. What the project will deliver, and what is new

By Month 36 the project delivers:

1. A coastal motion atlas and the relative sea level at each asset, from Sentinel-1 InSAR since 2014, corner reflectors and GNSS (D1.1).
2. A monitoring network chosen for the warnings and handed over to the operators (D4.2): piezometers with conductivity loggers, a barometer and sealed hub channels that correct them, a local rain gauge, GNSS, seismic nodes and, where access allows, fibre-optic sensing. Each sensor is bought only if its guaranteed value (band removed, years of safe operation added) is positive.
3. One calibrated model that couples groundwater flow and salt, aquifer compaction, liquefaction under a moving water table and coastal water levels (D2.1), and a fast neural-operator copy with a certified error bound (D2.2); a generative copy explores long storylines. The inputs and boundaries nobody measures are reconstructed from rain and heads, with bands valid when the simulator or rain product is wrong or the climate drifts.
4. Certified alarms for seawater intrusion, liquefaction and extreme water levels whose miss rate is bounded in advance (indicatively 5–10%), with a joint guarantee for the three assets, back-tested since 2014 under a pre-registered protocol (D3.1, D3.2).
5. Decisions for the water utility: the effect of each wellfield's pumping on salinity with the hidden drought demand removed; the salinity under drier recharge; the certified safe operating window and the year it closes; safe pumping tests that sharpen it; and a pumping rule agreed with the utility.
6. Twelve months of live bulletins with the four end-users, every outcome logged and the miss rates verified (D5.1); operating procedures and a policy brief (D6.2); open software and data with a DOI (D6.1).

How this advances the state of the art. The alarms apply conformal and anytime-valid methods of 2023–2026 to coupled coastal hazards. Six contributions are new and were tested before the project. (i) Guarantees for quantities nobody measures: inputs and boundaries known only through rain, calibrated on simulated aquifers and measured anchor periods, with a computable floor on how narrow each band can be. (ii) The value of a sensor as the guaranteed band it removes and the years of safe operation it adds (in tests, a tide gauge rated useful by a standard ensemble widened the band by 16–21%). (iii) Pumping effects free of hidden confounding: reading the recharge from the heads removed the bias of drought-driven pumping that history-based substitutes kept. (iv) The certified safe operating window and safe pumping tests chosen for the years they add. (v) A network that corrects its own sensors: reference channels remove air-pressure and logger-drift errors, and the robust design has a certified distance from the best network (+52% worst-case information in tests). (vi) Warnings for what happens between samples: the chance that chloride crosses its limit at any time in the season,

as a certified band over all models that fit the records (34 times the chance of an exceedance on a sampling date in tests). The integrated system reaches TRL 6 on site, with the operators.

5. How the project works

Seven work packages run over 36 months in three phases, separated by evidence gates with thresholds fixed in advance.

Observe and measure (Months 1–12). WP4 designs the network before buying it: each candidate sensor (piezometer with conductivity logger, rain gauge, GNSS, seismic node, barometer and sealed hub channels that correct the loggers for air pressure and drift) is ranked by the guaranteed warning band it removes and the years of safe operation it adds per euro; the network is chosen against the worst plausible sensor error, with a certified optimality gap (D4.1, Month 6); a logger or samples paired a few days apart are added where sparse samples leave crossings between them open. WP1 obtains vertical land motion and relative sea level at each asset from Sentinel-1 InSAR since 2014 and GNSS, and the head limit for pumping from aquifer compaction.

Model and certify (Months 4–30). WP2 builds one coupled model in stages (flow and salt, then compaction and liquefaction). The inputs and boundaries nobody measures are reconstructed from rain and heads by a generative prior with an ensemble update. Their bands are calibrated on simulated aquifers and on measured anchor periods, so they stay valid if the simulator is wrong; under drift the calibration is matched to the forecast horizon, and a computable floor shows how far each band is from the narrowest possible. Gate MS3 (Month 18) requires held-out heads, chloride, InSAR and seismic velocity inside the 90% intervals. A neural-operator copy with a certified error bound then serves the warnings. WP3 agrees thresholds and tolerable miss rates with the operators, builds the alarms (crossings between samples included, as a certified band over the models that fit the records), back-tests them since 2014 under a protocol registered before the results; gates MS4 (Month 22) and MS5 (Month 27) decide live operation and confirm the miss rates.

Operate, verify and hand over (Months 22–36). WP5 runs the system live for twelve months, logs every bulletin with its outcome, verifies the miss rates with anytime-valid tests and runs a post-event protocol. It estimates what each wellfield's pumping does to the salinity with the hidden drought demand removed: the heads, corrected for the measured drawdown, read the recharge, and a calibrated check rejects any substitute that misses a confounder. It computes the salinity under drier recharge, the certified safe operating window, its closing year and safe pumping tests that sharpen it, and agrees a pumping rule with the utility. WP6 trains early-career researchers and operators, releases open software and transfers the method to a partner coast. WP7 manages data access, security classification, risks and reporting.

Collaboration. The Albanian team leads WP1–WP5 and WP7; the partner-country team leads WP6, a transfer study on its own coast and a user-side review of the bulletins. Early-career researchers of both countries work across work packages, with two-month exchanges. End-users set thresholds, join a dry run and co-sign operational milestones; a steering committee with them meets every six months, and each gate has a named fallback.

6. The end-users: what they give and what they gain

End-user	Contribution (in kind; no SPS funds)	What it gains
Water utility: Shoqëria Rajonale Ujësjellës Kanalizime Durrës sh.a.	Access to production and monitoring wells for the loggers; records of heads, chloride and pumping under a data agreement; support for a yearly campaign of paired chloride samples; staff time for the chloride threshold, the pumping rule and safe pumping tests.	Certified intrusion warnings (also for crossings between samples), a joint pumping rule, the certified safe operating window, the monitoring network and trained staff.
Water resources agency: Agjencia e Menaxhimit të Burimeve Ujore (AMBU)	Access to monitoring wells under its control (existing wells only); records of levels, chemistry and abstraction under a data agreement; staff time for the warning threshold and the pumping and drought rule.	A monitoring design for its decisions, level and salinity sensors, a chloride early warning for each supply well, a pumping and drought rule, open software and training; no procurement, ownership or maintenance obligations.

End-user	Contribution (in kind; no SPS funds)	What it gains
Port: Autoriteti Portual Durrës	Access to agreed locations for sensors under its security rules; geotechnical records and quay design levels under a data agreement; staff time for the thresholds and the post-event protocol.	Liquefaction and extreme-water-level warnings for the port, post-event reports and trained staff.
Energy landfall operator	Access to agreed locations near the landfall under its security rules; design levels and geotechnical data; staff time for the thresholds.	Stability bulletins combining shaking, liquefaction, water level and subsidence.
Civil protection: Agjencia Kombëtare e Mbrojtjes Civile	A seat on the steering committee; the triggers of the post-event protocol and its co-design.	One picture of water, port and energy after earthquakes or storms; post-event reports and procedures.

The bulletins are advisory: operational decisions stay with the operators, and any warning to the public is issued only by the competent authorities. Data provided by end-users are restricted, used under data agreements, and every publication is reviewed for sensitive content before release; software and non-restricted data products are open, with a DOI.

7. Work plan and milestones

Work package	Title	Duration
WP1	Coastal motion and relative sea level	Months 1-36
WP2	One coupled model of the coast and its certified fast copy	Months 4-24
WP3	Certified multi-hazard alarm	Months 13-34
WP4	Monitoring network designed for the warnings	Months 1-36
WP5	Live demonstration at the wellfields, port and landfall	Months 22-36
WP6	Training, transfer and dissemination	Months 1-36
WP7	Management, data and risk	Months 1-36

Milestone	Month	What is achieved
MS1	6	Network design agreed
MS2	12	Network streaming; coastal motion known
MS3	18	Coupled model calibrated
MS4	22	Go/no-go for live operation
MS5	27	Miss-rate targets confirmed in back-testing
MS6	34	Twelve months of live operation completed
MS7	36	Hand-over to the operators

8. Budget (EUR)

Category	EUR
Equipment	182,800
Training	41,000
Communication and Publications	18,000
Travel	48,000
Consumables	22,700
Other	22,000
Stipends	48,000
Total requested from NATO SPS	382,500

Salaries, overheads and the in-kind contributions of the institutions and end-users are not charged to NATO. The monitoring network bought by the project is handed over to the operators.

9. Evidence from the preparatory work (2026)

- Boundaries known only through rain: on 500 simulated aquifers with four loggers, the raw ensemble missed 17–20% of the weeks; the calibrated bands met the 10% target (0.095–0.100).
- Pumping and hidden drought demand: in a test bed with eight wells, the summed effect of pumping on salinity (true value 0.94) was overstated by 0.59 without adjustment, by 0.56 with a substitute built from the pumping history (as in the Time Series Deconfounder) and by 0.00 with the recharge read from the heads; the physics check had size 6% and power 99–100%.

- Alarms and operating window: an alarm with an extreme-aware margin kept the share of seasons above the chloride limit at 4–9% across storylines, where a static alarm allowed 13–21%; the certified safe operating window closed after 38 years, while plans that watch one hazard believed they had 43–97 years; safe pumping tests chosen for the years they add raised the certified closing year from 10.8 to 31.9 years (true value 45) with chloride below 123 mg/L in 95% of runs; the anytime-valid test wrongly flagged a correctly calibrated alarm in 2.2% of runs, against 68% for re-checking a 95% bound after every event.
- Sensors that interfere: a 12-sensor network with a barometer and hub reference channels kept 52% more worst-case information than the classical design, certified within 18% of the best; ignoring the interference let 90% intervals cover only 49–62% of cases, the correction 88–93%.
- Crossings between samples: chloride with an 8-day time scale crossed 250 mg/L within a season with probability 0.78, 34 times the exceedance on a sampling date; quarterly samples left this anywhere in 0.18–1, a daily logger in 0.76–0.80; a conformal margin restored 90.3% coverage for skewed chloride (74% without).

These are tests on synthetic coasts and on the existing chloride model with illustrative values, not results at Durrës; the project calibrates and verifies them on site.

10. Contact

Assoc. Prof. Dr. Dulian Zeqiraj, NATO-country Project Director - Department of Energy Resources, Faculty of Geology and Mining, Polytechnic University of Tirana, Albania - dulian.zeqiraj@fgjm.edu.al